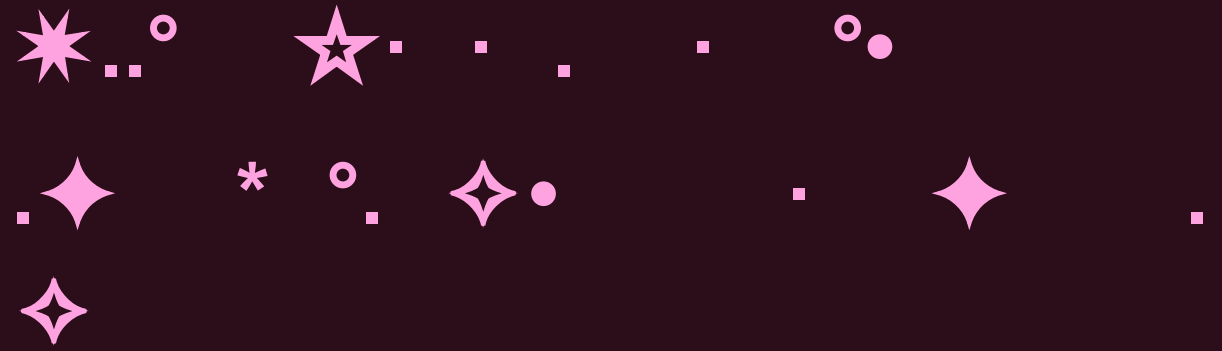


Convivial Coding for beginners with p5.js

Resources/tutorials:

nynnechristoffersen.com/files

Agenda ·°·* ✿✿❀✿✿❀

- Hi intro welcome
- Creative coding with. P5.js
- Snack break
- Exercises
- Round up + Resources to keep learning



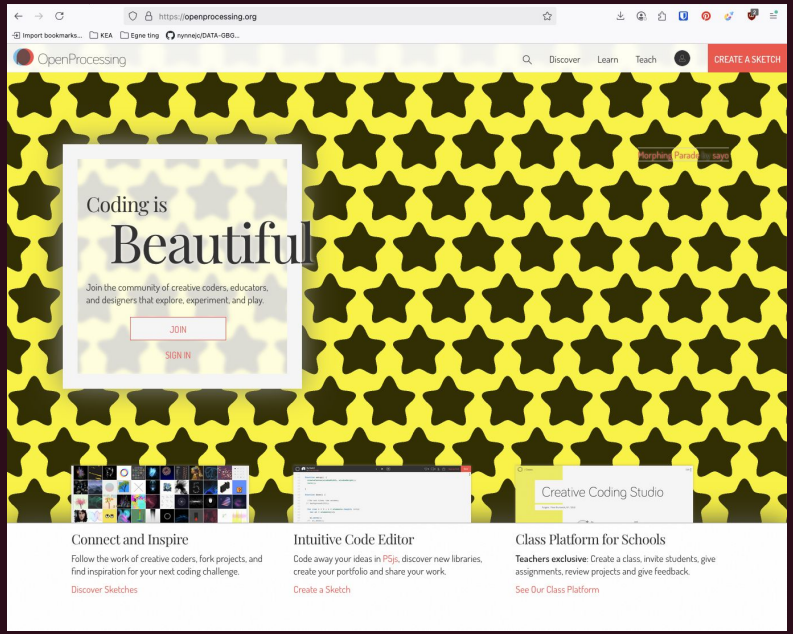
Hi my name is Nynne,
what's yours?

p5.js

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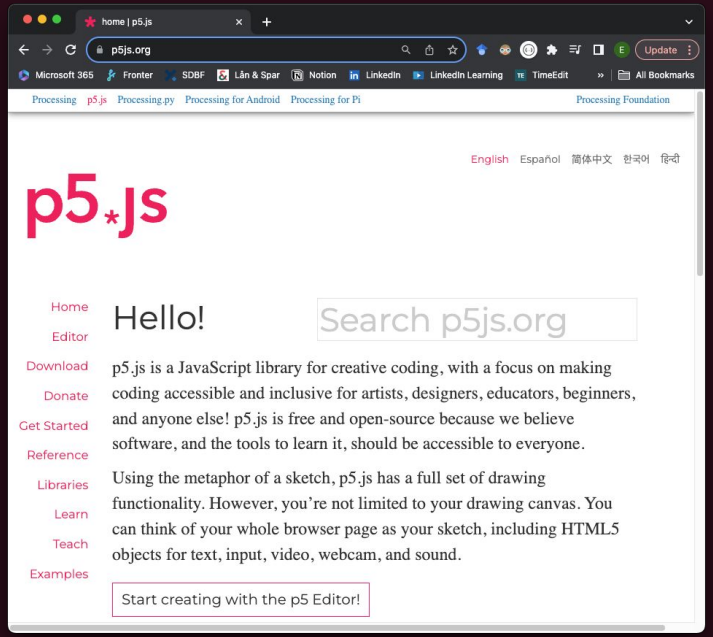
p5.js

- p5.js is a library that lets you draw graphic shapes and interactive graphic design and generative art for the screen, with code.
- Creative Coding

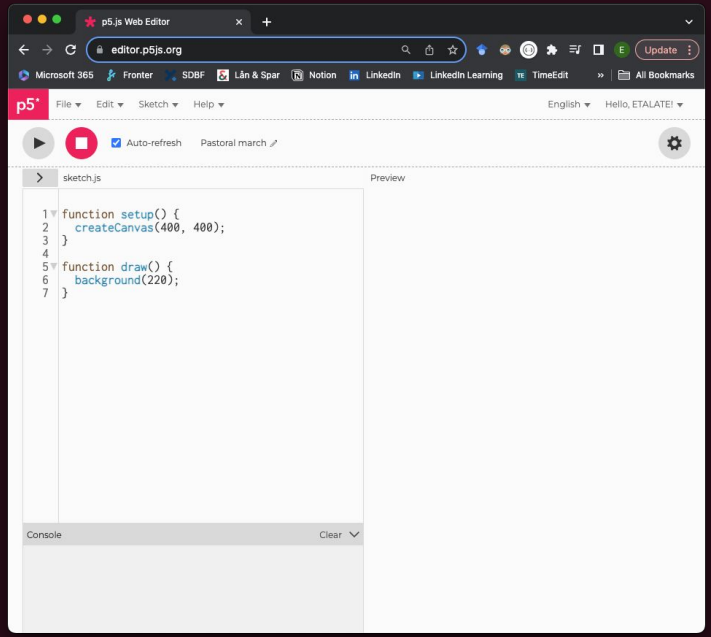


- https://openprocessing.org/
- https://exp.paperdove.com/chillin/

p5.js



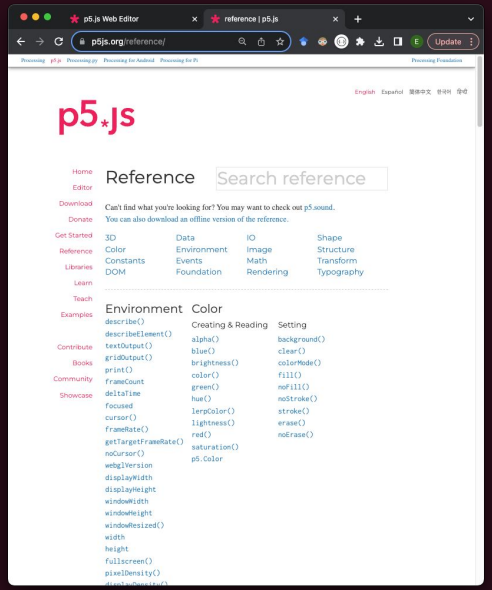
- https://p5js.org/



- https://editor.p5js.org/
- You must login to use it!

Our first sketch

- **setup()**
  - Is called as the first thing when the program runs. Usually contains start variables and functions e.g. Background color and screen size. Can only be called once.
- **createCanvas()**
  - Called only once. Must be placed in the **setup()** function. Creates a canvas with dimensions in pixels. Without parameters, canvas becomes 100x100 pixels.
- **draw()**
  - Runs the code inserted in the body in a loop until **noLoop()** is called.
- **background()**
  - Sets the background color. If not used, the background is transparent. A number is understood as grayscale and string as CSS color. Three parameters RGB (default), HSB or HSL.



- https://p5js.org/reference/

Our First Sketch: Shapes



Shapes

- **line()**
  - Syntax: line(x1, y1, x2, y2)
    - Draws a line
    - Has no weight / size and therefore cannot use fill(). Instead, use stroke() to change the color and strokeWeight() to change the width of the line.
- **circle()**
  - Syntax: circle(x, y, d)
    - Draws a circle
    - The first two parameters define the location of the center of the circle. Third parameter defines the diameter of the circle.
    - Alternative to ellipse()
- **square()**
  - Syntax: square(x, y, d)
    - Draws a square
    - The first two parameters define the location of the square's upper left corner. Third parameter defines the length of the side of the square.
    - Alternative to rect()



live coding

Line, circle & square

Exercise

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Solution for the exercise□□□□□



Snack Break

□□ Movement and Interactivity

Movement and Interactivity

- How do we make something move?
- Re-positioning our form
- Using our draw() function's loop

Movement and Interactivity

- **How do we make something interactive?**
- Take input from mouse or keyboard
- Use built-in I/O functions and variables in p5.js, for example mouseIsPressed

live Code

Moving rectangle

Exercise

✧･ﾟ: *✧･ﾟ:*

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Solution for the exercise



Snack Break

ADVANCED

Adding a sketch to a website

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Text editor ≫¦≪ ◎ï □ □ □.□ ˙ ₊

For this exercise you'll need a code text editor, if you don't already have one:

- Try Sublime
- Or VSCode

Our files ₊˚⊹♡✶★.˚

- index.html
- style.css
- sketch.js

Index.html ○✵

```
<!DOCTYPE html>
<html>

<head>
   <title>Creative Coding Yay!</title>
   <link rel="stylesheet" href="./style.css">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Softer Creative Coding Course</title>
   <script src="https://cdn.jsdelivr.net/npm/p5@2.0.0/lib/p5.min.js"></script>
</head>

<body>
   <h1>Hey hi!</h1>
   <h2>Look at my beautiful drawing</h2>
   <script src="sketch.js"></script>
</body>

</html>
```

style.css

```css
body, html {
 margin: 0;
 padding: 0;
 height: 100%;
 overflow: hidden;
}

canvas {
 position: fixed;
 top: 0;
 left: 0;
 z-index: -1;
}
```

sketch.js



In here we will add code from our favorite sketch



Thank you!

Happy Coding p5js Tutorials

The Coding Train Youtube